



Human experimental exposure to glyphosate and biomonitoring of young Swedish adults

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ABSTRACT

Glyphosate (GLY), N-(phosphonomethyl) glycine, is the most widely used herbicide in the world. It is a broad-spectrum herbicide, also used in crop desiccation. Agricultural workers may be occupationally exposed and general populations may be exposed to GLY mainly through diet. We studied the kinetics of GLY by measuring the parent compound and its metabolite aminomethylphosphonic acid (AMPA) in urine samples of three volunteers after an experimental oral exposure. We further examined GLY exposure by measuring GLY and AMPA in spot urine samples of 197 young adults in the general population in Scania, southern Sweden. Urine samples were analyzed using LC-MS/MS. In the experimental exposure, three healthy volunteers received an oral dose equivalent to 50% of the ADI for GLY. Urinary samples were collected up to 100 h after the exposure. The excretion of GLY to urine seemed to follow first-order kinetics and a two-phase excretion. The excretion half-life of GLY (density adjusted) was 6–9 h in the rapid phase and 18–33 h in the slower phase. The total dose recovered as unchanged GLY in the urine samples of volunteers was 1–6%. The metabolite AMPA was found to be 0.01–0.04% of the total dose of GLY. In the population of young adults, the median concentration was below 0.1 µg/L and a maximum concentration being 3.39 µg/L (density adjusted). AMPA was generally detected in lower concentrations (maximum = 0.99 µg/L). A moderate correlation (Spearman's $\rho = 0.56$) was observed between GLY and AMPA concentrations. Overall, the results may suggest that GLY and AMPA partly originate from separate exposures and that unchanged GLY is a more suitable biomarker of exposure.

1. Introduction

Although glyphosate is currently the most widely used herbicide (Connolly et al., 2020) and an important topic of discussion worldwide, little is known about the human toxicology, metabolism, and population exposure. Glyphosate (GLY), N-(phosphonomethyl) glycine, a broad-spectrum herbicide also used for crop desiccation, is the active ingredient in several commercial products, such as Roundup®. Since GLY is extensively used in agricultural food production, populations may be directly exposed in occupational settings or environmentally exposed through residues in food. GLY has previously been considered to pose a relatively low risk both for the environment and for the health of non-target species such as mammals (Williams et al., 2000). However, some recent studies report indications of associations with adverse outcomes in soil ecosystems, animals, and humans (Myers et al., 2016). Possible adverse health effects include carcinogenic, endocrine-disruptive, and microbiome-disruptive effects (Davoren et al., 2018; Kogenivas, 2019; Zhang et al., 2019). Furthermore, some studies have raised concerns over the possible health effects of other ingredients and/or mixture effects in commercial formulations of GLY-based

herbicides and over the toxicity of the GLY metabolite aminomethylphosphonic acid (AMPA) (Davoren et al., 2018). The current risk classifications and regulations are contradictory, with WHO—International Agency for Research on Cancer (WHO—IARC) classifying GLY as “probably carcinogenic to humans”, whereas the European Food Safety Authority (EFSA), US Environmental Protection Agency (EPA), and European Chemicals Agency (ECHA) conclude that the scientific evidence regarding GLY does not indicate carcinogenic potential for humans (Kogenivas, 2019). Further, a re-evaluation for a renewed approval of glyphosate as an active substance in plant protection products after December 2022 is being assessed within the EU (European Commission, 2020).

EFSA has previously established an acute reference dose (ARfD) and an acceptable daily intake (ADI) of 0.5 mg per kg of body weight for GLY (EFSA, 2015). Data from animal studies suggest that GLY is poorly bio-transformed and rapidly excreted unchanged, mainly in feces, but around 10–30% in urine (JMPR, 2016). Small amounts of the metabolite AMPA were reported in rats, from the microbial degradation of GLY after oral administration (Anadon et al., 2009).

The excretion of GLY in animals is reported to be two-phased, with a

Abbreviations: ADI, acceptable daily intake; AMPA, aminomethyl phosphonic acid; LC-MS/MS, liquid chromatography triple quadrupole mass spectrometry; LOD, limit of detection; GLY, N-(phosphonomethyl) glycine; QC, quality control; UER, urinary excretion rate.

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half-life of around 6 h for the shorter phase and 79–106 h for the longer phase (Williams et al., 2000). Information about the elimination kinetics of GLY in humans is very limited, but a recent oral exposure study reported a half-life of 9 h for GLY (Zoller et al., 2020) and another study of horticulturists reported the half-life in the range of 5.5–10 h (Connolly et al., 2019). Some biomonitoring studies assessing GLY exposure are available in the literature (Acquavella et al., 2004; Connolly et al., 2017; Conrad et al., 2017; Knudsen et al., 2017; Kongtip et al., 2017; Nova et al., 2020; Zhang et al., 2020), but human data on glyphosate are surprisingly scarce. Available European biomonitoring studies on GLY in non-occupationally exposed populations have mainly reported low concentrations, with median concentrations below the limit of detection (LOD) (Conrad et al., 2017; Connolly et al., 2018a,b; Soukup 2020). Occupationally exposed groups in several countries have reported higher concentrations ranging from a geometric mean at 0.7 µg/L in Irish horticulturists (Connolly et al., 2017) and up to 292 µg/L (Zhang et al., 2020) in Chinese workers within GLY production. With such few studies available, there is insufficient evidence to indicate that GLY is metabolized to AMPA in humans, or whether AMPA concentrations measured in human samples mainly derive from AMPA residues in cultivated food products (Connolly et al., 2020; Soukup et al., 2020). More information on the kinetics and fate of GLY in humans is required to support the interpretation of biomonitoring data, to relate exposure concentrations to currently existing health-based guidance values and to perform robust risk assessments (Connolly et al., 2020).

We aimed to study the excretion of GLY and AMPA in urine by conducting an experimental oral exposure in humans. Furthermore, we aimed to determine the concentrations of GLY and AMPA in urine samples of 197 young adults from a general population in southern Sweden.

2. Materials and methods

2.1. Experimental oral exposure

GLY was dissolved in 10 mL of Milli-Q water to a concentration of 10 mg/mL, which was used as a stock solution. Three healthy volunteers (one female and two males) received a single oral dose of GLY, equivalent to 50% of the ADI diluted in 250 mL of Milli-Q water. The volunteers fasted for 2 h before and after the oral dose and avoided conventionally grown commodities during the experiment. A pre-exposure urine sample was collected, and all voided urine was collected *ad libitum* up to 100 h after the oral dose. The times of voiding and the total volume of each sample were registered. All the samples were stored at −20 °C until analysis.

2.2. Analysis

The urine samples were analyzed using a method presented by Jensen et al. (2016), with minor modifications. The urine samples, calibration standards, quality controls (QCs), and chemical blanks (prepared in Milli-Q water) were added with 2% formic acid and internal standard. The samples were prepared in 96-well plates and analyzed using liquid chromatography triple quadrupole mass spectrometry (LC-MS/MS; QTRAP 5500, AB Sciex, Foster City, CA, USA). The limit of detection (LOD) was determined as three times the standard deviation of the GLY concentration in blank urine samples as 0.1 µg/L for both GLY and AMPA. A detailed method description including validation data is presented in the supplementary material. The laboratory participates in the German External Quality Assurance Scheme (G-EQUAS) for GLY analysis coordinated by the University of Erlangen-Nuremberg, Germany, with good results (Supplementary material).

2.3. Adjustment for urinary dilution

Creatinine concentrations in the urine samples were determined using an enzymatic method (Mazzachi et al., 2000) and were analyzed at the Department of Clinical Chemistry, Skåne University Hospital, Lund, Sweden. The laboratory is accredited for creatinine analysis. The concentrations of GLY were then divided by the creatinine concentrations in the samples to adjust for differences in urinary dilution. Additionally, the density of the urine samples was measured in our laboratory using a hand-held refractometer. The adjustment for urinary density, C_d , was calculated according to $C_d = C_{\text{observed}} \times (1 - \rho_{\text{mean}}) / (1 - \rho_{\text{sample density}})$, where C_{observed} is the concentration of the analyte in the urine sample, ρ_{mean} is the mean urine density for the study participants, and $\rho_{\text{sample density}}$ is the density of the urine sample. The mean urine density of the volunteers (1.015) was used for density adjustment in the oral exposure experiment, and the mean urine density of the population (1.019) was used for adjustment in the population biomonitoring. Furthermore, full void volume was measured for each sample in the oral exposure experiment and was used to estimate the urinary excretion rate (UER).

2.4. Estimation of elimination half-life

The elimination half-life was estimated using the quantified concentrations of GLY in the urine samples obtained from the volunteers in the oral experimental exposure. The excretion half-lives were calculated from the slope of the curve obtained from the plot of the natural log-linear concentrations versus mid-time points. Estimations were made using unadjusted data and for concentrations adjusted for density, creatinine, and urinary excretion rate (UER).

2.5. Biomonitoring of GLY in young adults

A spot urine sample was collected from each of 197 young adults (109 females and 88 males) in 2017 to assess environmental exposure to GLY through analyses of GLY and AMPA. The recruitment was performed at three schools in Scania, southern Sweden. The participants were aged 18–19 years and were in their last year of secondary school. They were informed and recruited at their schools, where we also collected the samples. The recruitment was originally performed for a national biomonitoring program organized by the Swedish Environmental Protection Agency (EPA), to assess exposure to environmental pollutants in the population (Noren et al., 2020).

2.6. Ethical approval

The human experimental exposure studies and the biomonitoring of the exposure to GLY in the studied population in southern Sweden were ethically approved by the Regional Ethical Review Board, Lund University, Sweden (Dnr 463/2005, Dnr 2010/41, Dnr 2010/465, and Dnr 2013/6). The volunteers and the participants in the population study gave written informed consent to participate.

3. Results

3.1. Experimental oral exposure

After oral administration of GLY, the urinary concentrations of GLY increased rapidly and C_{max} was reached within 2 h (Table 1). The measured GLY concentrations were 3–1400 µg/L. The dose recovery was 1–4% within the first 24 h, and the total dose recovery up to 100 h after exposure was 1–6% for the different volunteers. The C_{max} of AMPA was reached 5–6 h after exposure (Table 1). The measured AMPA concentrations were low, between the LOD and 6.4 µg/L, and the total dose recovery was 0.01–0.04% of the administered GLY dose. The concentrations of GLY and AMPA in the pre-exposure samples were below the LOD of 0.1 in all volunteers. The freshly made solution of 200 µg/L of the

Table 1

Descriptive characteristics of the study volunteers, the oral doses of GLY and the urinary concentrations before and after the exposure (density adjusted). Recovery of the dose as unchanged GLY and the metabolite AMPA is presented in percentages after 24 h and total dose recovery. $C_{\max}$ is the maximum concentration (density adjusted) excreted in the urine, and estimated $T_{\max}$ is the mid-time point of maximum excretion.

Volunteers	Age (years)	Weight (kg)	BMI	Oral dose ^a (mg)	Pre-exposure		GLY				AMPA		
					GLY (µg/ L)	AMPA (µg/ L)	Recovery (%)		Cmax (µg/ L)	Tmax (h)	Recovery total (%)	Cmax (µg/ L)	Tmax (h)
Female	73	57	23	14.3	<LOD	<LOD	1.2	1.9	234	2	0.02	1.7	6
Male 1	48	75	22	18.8	<LOD	<LOD	0.9	1.1	197	2	0.01	0.8	5
Male 2	42	59	20	14.8	<LOD	<LOD	4.2	6.3	1014	2	0.04	6.4	6

^a The oral dose of GLY was 50% of the ADI and was calculated according to the body weight of each participant.

GLY formula administered to the volunteers was analyzed for AMPA residues, which were found to be below LOD.

The urinary elimination of GLY seemed to follow first-order kinetics and a two-phase excretion, with an initial rapid phase between 6 and 9 h followed by a slower phase between 18 and 33 h. Fig. 1 shows the elimination curves of GLY after the exposure, with unadjusted concentrations and concentrations adjusted for creatinine, density, and UER. The normal scale unadjusted excretion plot of GLY is available in the supplementary material (Fig. S2). The urine concentrations did not return to the pre-exposure values even after 80–100 h (Fig. S2). AMPA elimination seemed to follow a similar two-phase elimination curve but was more difficult to determine because of the low concentrations and

great variability between the volunteers. The density-adjusted, creatinine-adjusted, and unadjusted kinetic plots for AMPA are shown in the supplementary material (Fig. S1A–C). The estimated excretion half-lives of GLY and AMPA are shown in Table 2.

3.2. Biomonitoring of GLY in young adults

In this population, GLY was found in detectable concentrations above the LOD of 0.1 $\mu\text{g}/\text{L}$ in 20% of the urine samples, and AMPA was found in concentrations above the LOD of 0.1 $\mu\text{g}/\text{L}$ in 29% of the samples. The median and mean concentrations of GLY were both below LOD. The maximum concentration (density adjusted) was 3.39 $\mu\text{g}/\text{L}$ for

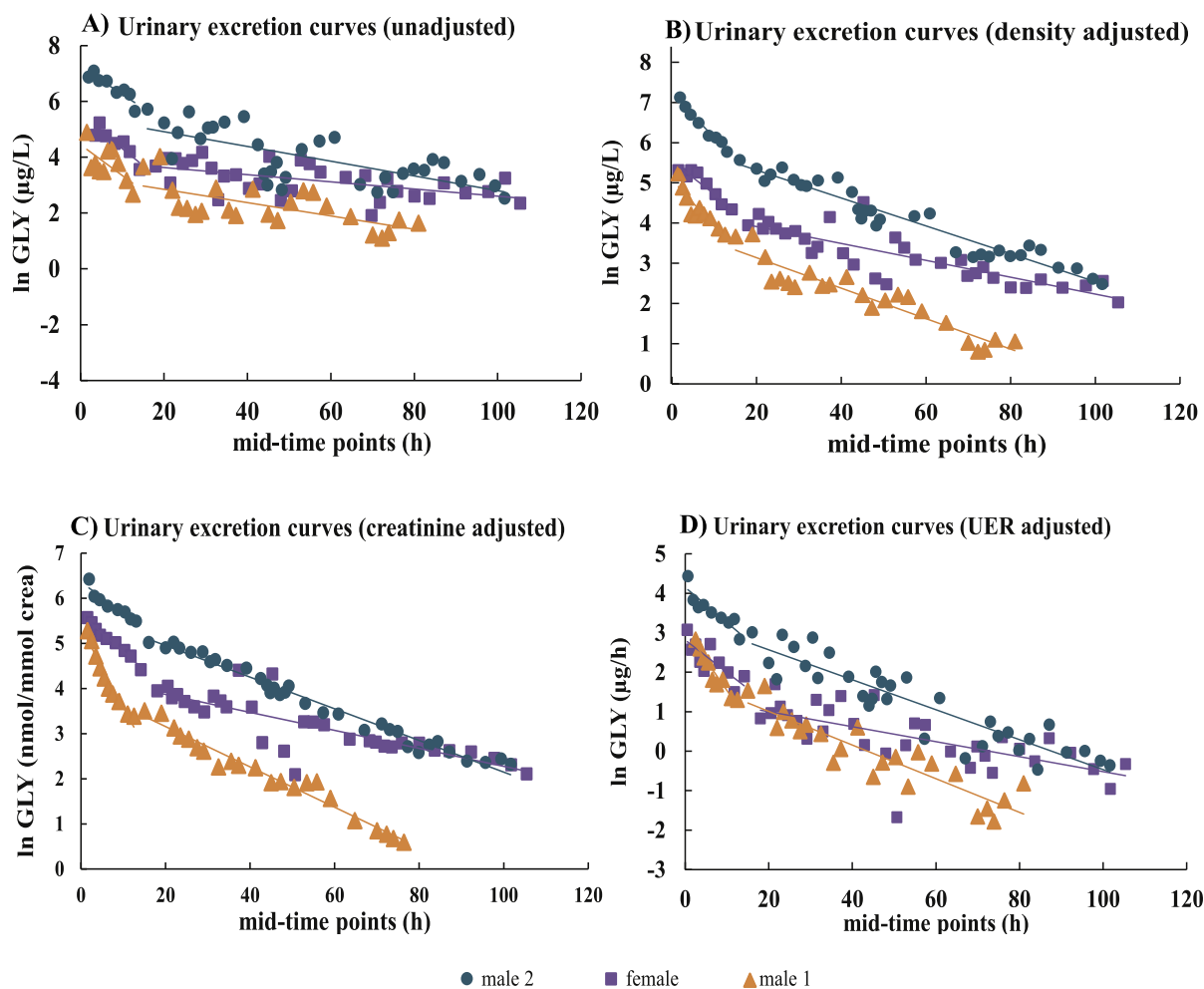


Fig. 1. Graphs of urinary excretion curves for GLY (parent compound) for the three volunteers after a single oral dose of GLY. The graphs show ln-transformed: A) unadjusted concentrations ($\mu\text{g}/\text{L}$), B) density-adjusted concentrations ($\mu\text{g}/\text{L}$), C) creatinine-adjusted concentrations (nmol/mmol creatinine), and D) UER-adjusted concentrations ($\mu\text{g}/\text{h}$) plotted against the mid-time points of sample collection.

Table 2

Estimated excretion half-lives of GLY in urine samples of study volunteers. Data on two-phase excretion half-lives (h) in urine are presented as unadjusted, density-adjusted, creatinine-adjusted, and UER-adjusted data with the correlation coefficients (*r*) for the short and long excretion phases, separately.

Compound	Volunteer	Excretion phase	<i>t</i> _{1/2} , unadjusted (h)	<i>r</i>	<i>t</i> _{1/2} , density adjusted (h)	<i>r</i>	<i>t</i> _{1/2} , creatinine adjusted (h)	<i>r</i>	<i>t</i> _{1/2} , UER adjusted (h)	<i>r</i>
GLY	Female	short	5.7	0.82	8.6	0.89	9.4	0.98	8.5	0.65
		long	54	0.33	33	0.70	35	0.67	37	0.45
	Male 1	short	6.3	0.42	6.0	0.83	3.9	0.94	4.5	0.92
		long	29	0.45	18	0.88	16	0.97	16	0.82
	Male 2	short	7.0	0.83	6.2	0.98	10	0.98	7.8	0.82
		long	26	0.51	20	0.57	20	0.57	18	0.83
	Female	short	4.6	0.88	4.4	0.99	4.8	0.94	–	–
		long	173	0.02	56	0.26	62	0.22	–	–
AMPA	Male 1	short	61	0.02	14	0.77	10	0.63	–	–
		long ^a	–	–	–	–	–	–	–	–
	Male 2	short	3.3	0.92	3.6	0.97	3.8	0.99	–	–
		long	50	0.26	29	0.88	27	0.83	–	–

^a Concentrations for the long excretion phase of Male 1 were <LOD.

GLY in urine and 0.99 µg/L for AMPA. Urinary concentrations are summarized in Table 3. The Spearman rank correlation indicated a significant, but only moderate, correlation ($\rho = 0.56$) between GLY and AMPA in the population.

4. Discussion

GLY is the most widely applied herbicide worldwide, and there is growing scientific and public interest in its effects on both human health and the environment. It is therefore noteworthy that there are not more available studies of GLY exposure, health effects, and fate in humans. The high polarity and amphoteric nature of GLY make it a difficult compound to analyze, and few publications address the issue of measurements in biological fluids.

This is one of few current studies reporting on oral exposure experiments with GLY in humans, using a known dose of 50% of ADI and observing the urinary excretion up to 100 h. The experiment was limited to three volunteers but may still be of value for the interpretation of GLY exposure when biomonitoring populations or intoxication cases. We found a low recovery of GLY in the urine of the three volunteers, in good agreement with a recent study by Zoller et al. (2020). They suggested a total recovery of 1% based on a study with 12 participants exposed through food, in which the GLY and AMPA doses were quantified in the food samples. Both results are in agreement that the elimination of GLY in urine is much lower in humans than in animals, as animal studies show that more than 20% of administered GLY concentrations were recovered in urine (Anadon et al., 2009; Brewster et al., 1991; Williams et al., 2000). Furthermore, the recovery of AMPA in our study was in line with the results of Zoller et al. (2020), who reported recovery of 0.09–0.28% of AMPA related to the administered dose of GLY. However, relative to the dose of AMPA detected in food samples, 10–33% of AMPA was recovered in urine by Zoller et al. (2020). Whether the lower urinary excretion rates of glyphosate in humans also indicate a lower absorption/resorption, and lower systemic availability, in humans warrants further investigation in future research.

In our observations, the urinary elimination appeared to be two-phased, similar to the observations reported in animal studies. One

study reported an initial excretion phase at 6 h and a second phase between 79 and 106 h (Williams et al., 2000), another observed a half-life of 14 h in plasma after oral administration in rats (Anadon et al., 2009), and yet another reported a half-life of 8–11 h in the plasma of rats after oral administration by gavage (FAO/WHO 2017). Two recent studies in humans reported a half-life for GLY of 9 h after urine collection up to 48 h (Zoller et al., 2020) and an average half-life in the range of 5.5–10 h when calculations were based on different urinary adjustment metrics (Connolly et al., 2019). The results from the rapid excretion phase in our study are very similar to these findings. Further, Connolly et al. (2018a) reported that *C*_{max} was reached in horticulturists between 1 and 3 h after the end of the work shift, and Zhang et al. (2020) reported that *C*_{max} was reached in workers involved in GLY production within 1 h after the shift ended. This is in line with our results where *C*_{max} was reached at 2 h after the oral dose, but the different routes of exposure should be considered in this comparison. The urine collection period and exposure dose might affect the observed excretion pattern, in which a two-phase excretion could have been missed due to a shorter follow-up or a lower exposure dose. However, only a small fraction of the recovered GLY concentration was excreted after 24 h. There seem to be some inter-individual differences in GLY excretion observable in the three volunteers, even though they were administered a GLY dose related to each person's weight. The concentrations of both GLY and AMPA were below LOD in the pre-exposure urine samples from the volunteers. Since our data is limited to three individuals, no conclusions can be drawn on why the recovery and maximum concentration were much higher in Male 2 compared to the other two participants. The excretion patterns of all volunteers, including Male 2, in Fig. 1 does not suggest any additional exposure during the experiment. The estimated half-life of AMPA appeared to be two-phased, but its concentrations were close to the LOD with great variability between the volunteers.

The urinary concentrations and detection frequency of GLY and AMPA in the population of Swedish young adults were low. Comparing the highest concentration in this population with the *C*_{max} in the experimentally exposed volunteers shows that GLY concentrations in the population are far below the concentrations from an exposure equivalent to 50% of the ADI. Overall, the urinary concentrations are in line with previously reported results from other countries. Danish school children and mothers had detectable concentrations with means of 1.96 and 1.28 µg/L, respectively (Knudsen et al., 2017). In a time-trend study (2001–2015) of German adults, median concentrations ranged from below LOQ to 0.18 µg/L in 24-h urine samples in different years (Conrad et al., 2017). Another German study of an adult population found detectable concentrations above LOQ only in 8.3% of the population (Soukup et al., 2020). A study of adults from 18 European countries presented a mean GLY concentration of 0.21 µg/L (Hoppe et al., 2013). A recent study from Portugal of adults from the general population volunteering to participate reported an arithmetic mean concentration

Table 3

Urinary concentrations adjusted for density (µg/L) and creatinine (nmol/mmol creatinine), and detection frequency of GLY and AMPA in a Swedish population of young adults.

Biomarker	LOD	%>LOD	N	Median	Mean	95th P	Max
GLY (dens)	0.1	20	196	<LOD	<LOD	0.24	3.39
GLY (crea)			194 ^a	<LOD	<LOD	<LOD	1.10
AMPA (dens)	0.1	29	195	<LOD	<LOD	0.25	0.99
AMPA (crea)			193 ^a	<LOD	<LOD	0.16	0.53

^a Two of the samples did not contain enough volume for analysis of creatinine.

of GLY at 0.05 µg/L and 0.1 µg/L in urine from two different test rounds, and 0.08 µg/L and 0.1 µg/L of AMPA (Nova et al., 2020). An occupationally exposed group of Irish horticulturists had a mean GLY concentration of 0.66 µg/L in post-work samples (Connolly et al., 2017). A recent study from China reported median urinary concentrations of GLY in the range of 67–3297 µg/L and AMPA 22–388 µg/L in a population working with glyphosate production, reported to be exposed through inhalation (Zhang et al., 2020). A review of population exposures to GLY observed arithmetic mean concentrations in urine of 0.16–7.6 µg/L (Gillezeau et al., 2019). Despite the differences in analytical methods for GLY analysis and sample collection procedures, the reported concentrations in Europe are in good agreement with the low concentrations and detection frequencies of GLY and/or AMPA in urine samples found in different populations. Furthermore, we observed a moderate correlation between AMPA and GLY concentrations in the samples from the general population of Swedish young adults. GLY and AMPA seemed to have similar elimination half-lives and kinetics based on observations from our oral exposure assessment with a slightly different T_{max} and recovery. If GLY is metabolized to AMPA in humans, they should be detected and correlate to some extent in spot urine samples after exposure to GLY. However, possible inter-individual differences in metabolism and/or elimination should still be considered. Other studies have also found a moderate or weak correlation between GLY and AMPA concentrations (Conrad et al., 2017; EFSA, 2015; Nova et al., 2020; Soukup et al., 2020). It is plausible that GLY is metabolized to AMPA by soil microbiota before entering the body via diet, resulting in different exposure sources. Still, a small fraction of the AMPA measured in urine is attributable to the metabolism of GLY based on our observations in the oral exposure experiment. An animal study suggests that the metabolism of GLY to AMPA mainly is the result of gut microflora activity (Brewster et al., 1991).

Overall, the available results suggest that the absorption and elimination patterns of GLY after exposure differ between human and animal studies, with sufficient human data being lacking. A few animal studies have assessed distribution and uptake and compared different routes of exposure. A recent study from China assessed occupational exposure to GLY in indoor air, where the highest measured median concentration range was 3–12 mg/m³ (Zhang et al., 2020). The studied workers had median concentrations of GLY in urine ranging from 292 to >3000 µg/L. In the present study, we administered an oral dose of 14–19 mg, resulting in urinary C_{max} values of 200–1000 µg/L. If we assume a working day of 8 h, that the inhalation volume is 1 m³/h and a similar uptake of 1%, exposure routes can be crudely compared. Based on these assumptions, the estimated exposure dose from indoor air would be 24–100 mg of GLY when the reported urinary C_{max} was 2000–8000 µg/L. This comparison indicates that GLY uptake is similar for oral uptake and inhalation, with possibly a slightly greater uptake from inhalation. There is a great need for studies of GLY that assess different exposure routes in humans to explore this matter further.

It should be considered that GLY may bind to metal cations due to its chelating property (Mertens et al., 2018). We used Milli-Q water as the medium for oral administration of the GLY dose in the exposure assessment to avoid possible complex formation or GLY interactions with ions. Variations between our results and those of Zoller et al. (2020) could be due to the different administration of the oral dose of GLY in Milli-Q water or through food products containing GLY residues.

Furthermore, only a few studies present measurements of GLY in human blood samples from acute intoxication cases. One study concluded that blood concentrations correlated well with ingested doses and were more useful for clinical evaluations than were urinary concentrations (Zouaoui et al., 2013). Further exposure assessments including measurements of GLY and AMPA in blood samples would advance our understanding of the absorption, distribution, and elimination of GLY in humans. More data are needed to investigate whether different exposure routes and differences in diet might affect the uptake, bioavailability, and/or elimination of GLY in humans.

4.1. Adjustment for urinary dilution

All urinary GLY concentrations adjusted for density or creatinine, or presented in terms of UER, gave better correlations (r) in the half-life plots than did the non-adjusted concentrations. Still, both creatinine and density are affected by the body composition of the individual and by diet, age, and sex, among other factors. However, density values are suggested to be less affected by some factors than are creatinine concentrations (Suwazono et al., 2005). In biomonitoring studies, in which the population composition is usually complex due to a wide age range, a particular sex ratio, and varied health status, density may be an appropriate choice for dilution correction, especially when comparing different groups such as men and women. However, density might also be affected by several other factors, including common diseases such as diabetes or kidney disorders (Chadha et al., 2001), and the measurement itself is difficult to automate. Dilution correction with UER is only an option when timed full-void urine samples are available. Overall, applying any of the methods for adjusting the urinary dilution should be preferred to using unadjusted urine concentrations.

5. Conclusions

In the human oral experimental exposure, the urinary excretion of GLY seemed to follow a two-phase excretion. The elimination half-lives (density adjusted) of GLY in urine were 6–8 h for the initial short excretion phase and 18–33 h for the longer second excretion phase. The recovery of the oral dose of GLY was 1–6% in the three volunteers; AMPA was found in trace concentrations of around 0.01–0.04% of the administered dose of parent GLY. In the Swedish population of young adults, the median and maximum GLY concentrations in urine were 0.03 and 3.39 µg/L, respectively; AMPA was generally found in lower concentrations. A moderate correlation between GLY and AMPA concentrations ($\rho = 0.56$) together with concentrations of AMPA in the same range as GLY in the population, as opposed to our experimental exposure, may suggest that the detected GLY and AMPA reflect separate exposures from different sources. The concentrations of GLY found in the Swedish population were consistent with the results of other European studies. Our observations indicate that the urinary concentrations in the studied population are far below the maximum concentrations found in our human exposure study, after intake of a dose equivalent to half of ADI (0.5 mg/kg body weight per day).

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijheh.2020.113657>.

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